Watch Your Position: Neural Inertial Localization With a Single Wrist-Worn Device

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***Abstract*—Wearable devices, such as smartwatches, have been widely used by the public for health monitoring, exercise tracking, and message alerts. Regarding localization applications, although smartwatches have been explored by many researchers, current solutions typically require combining data from multiple sensors or multiple mobile devices for reliable position estimation, which is inconvenient for practical usage. In this paper, we propose a novel indoor localization method that relies solely on an inertial measurement unit (IMU) of a smartwatch. First, we design a smartwatch-based neural inertial odometry (WNIO) that adapts to multiple motion patterns and achieves low-drift dead reckoning. Second, we detect interactions between the arm and the environment, such as opening a door, to formulate absolute position measurements that help constrain error drift. A multi-hypothesis Kalman filter (MHKF) is employed to ensure reliable matching and localization. We conducted real-world experiments to validate the effectiveness of the proposed method. The results demonstrate that our method achieves reliable dead reckoning across various motion patterns, including walking and running at different speeds. Furthermore, the method effectively mitigates error drift without relying on any external signals.**

***Index Terms*—Smartwatch, inertial odometry, indoor localization, deep learning, IMU.**

I. Introduction

NOWADAYS, an increasing number of people are wearing smartwatches for various applications [1], [2], [3], such as health monitoring, workout tracking, and message notifications. Compared to smartphones, smartwatches fit more closely to the body and are therefore often utilized for human activity recognition [4] or tracking arm posture in motion-based control, enabling more convenient interaction [5].

Beyond posture, accurate position estimation is also essential for enhancing cooperation between humans and machines, as well as their interaction with surrounding environments. For pedestrian localization, particularly in indoor settings, various techniques have been proposed [6], [7], including pedestrian dead reckoning (PDR), ranging, fingerprinting, and simultaneous localization and mapping (SLAM). Most of these methods are designed and tested for smartphones, as they typically integrate various sensors and offer sufficient computing power to support advanced localization algorithms. Although smartwatches and other wearable devices have gradually been incorporated into combined localization systems [8], these systems still rely on smartphones for reliable position estimation. This reliance can be attributed to two main reasons. First, wearable devices typically include only a limited set of sensors, which makes it difficult to achieve localization performance comparable to that of smartphones. Second, although PDR using smartwatches has been studied, its performance often degrades due to the variability of arm movements, unlike smartphones, which are generally kept in a fixed or stable position. However, the need to carry smartphones can introduce additional burdens. On the one hand, users are required to carry redundant devices. On the other hand, smartphones must be either held in the hand or kept in a pocket, which can be inconvenient for certain applications such as running, health or child/elder monitoring, and rescue operations.

To address the issues mentioned above, this paper proposes a novel indoor localization method that relies solely on the inertial measurement unit (IMU) of a smartwatch. The main contributions of this work are summarized as follows:

- To the best of our knowledge, this is the first work to propose a neural inertial localization method that relies solely on smartwatch IMU data, without requiring any external devices or signals for localization.
- We develop a wearable neural inertial odometry (WNIO) approach that adapts to varying arm-swing motions and achieves low-drift dead reckoning. An adaptive error-state Kalman filter (ESKF) is developed to refine attitude estimation for IMU frame transformation, enabling more accurate velocity regression through a lightweight neural network and improving the robustness of the odometry.
- To further mitigate drift, we exploit human-environment interaction activities, such as door-opening, as additional localization constraints. A multi-hypothesis Kalman filter (MHKF) is proposed to resolve ambiguous activity associations, resulting in more accurate position estimation than traditional matching methods.

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The video (https://www.bilibili.com/video/BV17ihJzEEdh/?spm_id_from=333.1387.homepage.video_card.click&vd_source=621b8b6a8d3ffbbf4d5cdd2b3ab35347 BiliBili) is also shared to display our research.

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- Extensive real-world experiments are conducted to validate the proposed approach. Its performance is evaluated under various motion patterns and indoor scenarios.

The rest of the paper is structured as follows: In Section II, the relevant work is discussed. In Section III, the methodology overview of the proposed method is introduced. The proposed method is described in detail in Sections IV and V. In Section VI, experiments and discussions are given. Finally, Section VII concludes the paper.

II. Related Works

Indoor localization based on mobile devices has been widely studied for many years, resulting in the development of various advanced techniques. Among them, methods that exploit ambient signals and environmental features without requiring additional installations have attracted considerable attention, as they enable widespread and cost-effective deployment for the public.

Fingerprinting is a representative example, which typically leverages Wi-Fi, Bluetooth Low Energy (BLE), or magnetic field signatures to construct a fingerprint database [9], [10]. In the localization step, the position can be estimated by matching the measured features with the database. Although this technique yields satisfactory results, constructing the database is both labor-intensive and time-consuming. Furthermore, the need for regular updates further limits its practical applicability. Vision-based methods present another appealing approach, typically implemented via visual-inertial odometry [11] or image-based localization [12]. However, these methods often consume significant power, and creating a comprehensive image database is often unrealistic. By contrast, PDR does not require a dedicated database for localization, as it estimates the position by accumulating step lengths [13]. In addition, another advantage of this method is that it primarily relies on the built-in IMU (6-axis or 9-axis), which is low-cost and widely integrated into modern mobile devices.

Currently, extensive research has been conducted on PDR, which can be categorized into model-based and data-driven approaches. Classical model-based PDR algorithms typically consist of three main components: step detection, step length estimation, and heading estimation [13]. However, model-based methods can easily degrade when pedestrian motion patterns change or when the pose of the mobile device relative to the human body is unknown. Although researchers have proposed various techniques to address these issues [14], [15], significant errors may still occur, particularly when using mobile devices with complex motion patterns, such as smartwatches. Recently, data-driven PDR has attracted significant attention, as it can partially overcome the limitations of model-based methods. Chen proposed the IONet [16], which uses a long short-term memory (LSTM) [17] to regress changes in distance and heading from sequences of raw IMU data. It can overcome the shortcomings of model-based PDR, particularly in scenarios without periodic motion. Building on this work, Chen further developed light IONet (L-IONet) to reduce the computational load on mobile devices such as smartphones and smartwatches [18]. Herath introduced RONIN [19]. Multiple human motion scenarios, such as messaging, sitting, and taking photos, are considered. Three neural inertial navigation variants using smartphones are designed based on ResNet [20], LSTM, and temporal convolutional network (TCN) [21]. Cao proposed RIO [22], a rotation-equivariant supervised learning approach for training inertial odometry models. It reduces the dependence on large amounts of labeled data for model training. In addition, an adaptive test-time training (TTT) strategy is introduced to enhance the model's generalizability to unseen data. Rao introduced CTIN [23]. It combines a ResNet encoder for capturing spatial IMU context with an attention-based Transformer [24] decoder for temporal fusion, enhanced by uncertainty-aware multi-task learning to achieve accurate 2D velocity regression. In addition, unlike pure odometry, Herath proposed an absolute localization method using only inertial sensor measurements from smartphones [25]. Their approach employs a scene-specific, transformer-based neural architecture that maps a velocity sequence to a location likelihood distribution.

The data-driven PDR methods mentioned above mainly focus on smartphone applications. However, some have been adapted for wearable devices. For example, Liu presented TLIO [26], in which the data is collected and tested using a headset. TLIO regresses 3D displacement estimates along with their uncertainties. A stochastic cloning extended Kalman filter (EKF) is used to tightly fuse these relative state measurements with inertial navigation to estimate pose, velocity, and sensor biases. Yi proposed a real-time human 3D translation and pose estimation method [27] using six wearable inertial sensors. A supporting foot-based velocity estimation is combined with a recurrent neural network (RNN)-based approach to estimate global translations. Sun introduced Suite-IN [28], which aggregates motion data from the Apple suite, including iPhone, Apple Watch, and AirPods, to enhance global motion estimation. Although combining multiple devices can improve PDR performance, it may not be very efficient or easily scalable to a large number of users, who typically wear only one or two devices. In contrast, using a single wearable device remains highly appealing, as it naturally integrates with the body and enables position tracking without additional burden.

The smartwatch is undoubtedly one of the most ideal wearable devices for this purpose. Currently, some research has been conducted on human activity recognition using smartwatches [29], [30]. However, PDR based on a single smartwatch faces many challenges [31], mainly due to its more complex motion compared to smartphones. To address these issues, several studies have explored smartwatch-based PDR solutions. Lin proposed a convolutional neural network (CNN)-based PDR method specifically for smartwatches [32], where a global navigation satellite system (GNSS) is integrated to mitigate error drift. Park presented an adaptive detection method based on learning-based human activity recognition, which enhances step detection accuracy across different motion modes, thereby improving overall PDR performance [33]. To achieve reliable heading estimation, Park proposed a robust principal component analysis (PCA)-based approach [34] for estimating walking direction using smartwatches. Although these methods have

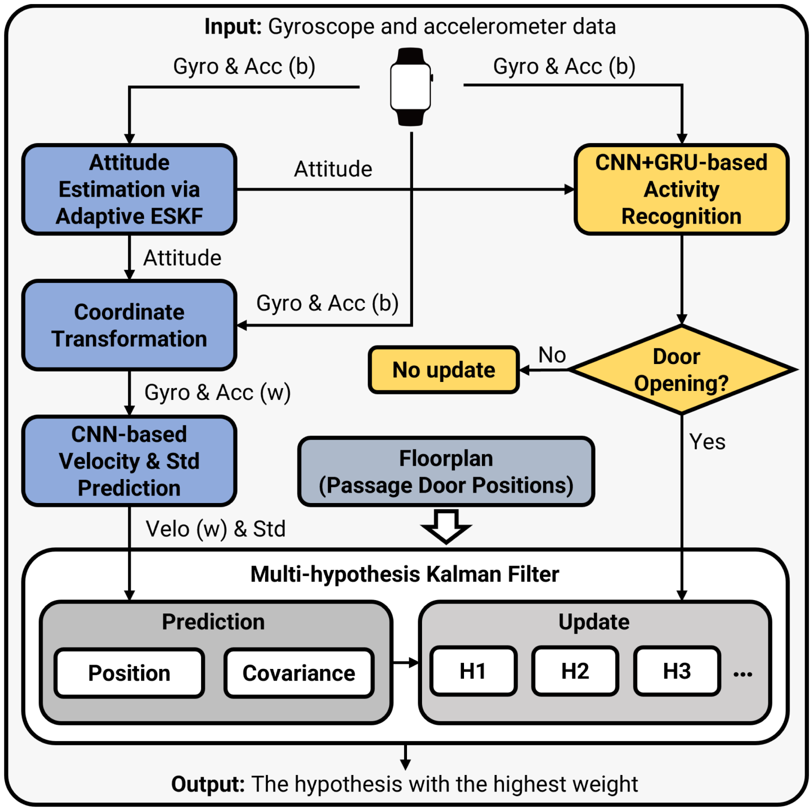


Fig. 1. The structure of the proposed method, in which the input is the 6-axis IMU measurements (gyroscope and accelerometer data) from a smartwatch. The output includes the hypothesis (position) with the highest weight.

achieved satisfactory improvements, they still suffer from the complex motion of the arm, which leads to degraded PDR performance in some cases.

III. Overview of the Proposed Method

The overall structure of the proposed method is illustrated in Fig. 1. The input consists of gyroscope and accelerometer data collected from a smartwatch. The proposed method comprises three main components.

First, the left part of the structure focuses on the smartwatch-based neural inertial odometry (WNIO). To reduce the impact of varying roll and pitch angles on inertial measurements, the gyroscope and accelerometer data are first transformed into a global reference frame (world frame) using reliable attitude estimates. In typical attitude and heading reference systems (AHRS), gyroscope integration captures rapid rotations, while accelerometer measurements are used to correct drift in roll and pitch under static or low-dynamic conditions. However, in smartwatch-based applications, significant linear accelerations during walking or running, as well as intermittent arm pauses, make attitude estimation particularly challenging. To address this, we introduce an innovation-based adaptive error-state Kalman filter (ESKF) that dynamically adjusts its correction gain based on motion state, thereby mitigating the influence of external accelerations on attitude estimation.

This filter acts as a physics-guided pre-processing module that projects raw IMU data from the body frame to a gravity-aligned world frame using physically consistent orientation estimates. By explicitly decoupling attitude estimation from the learning process, our framework eliminates the need for neural networks to learn rotation-invariant features, reducing model complexity and enhancing generalization across diverse motion patterns. Moreover, since ground-truth orientation is generally unavailable in wearable scenarios, the proposed approach enables label-free orientation preprocessing, leveraging the adaptive ESKF to provide robust and interpretable attitude estimates without requiring external supervision. After transforming the IMU data using the estimated attitude, a CNN-based model predicts the 2D velocity in the world frame along with its associated uncertainty.

In indoor environments, human activities are generally associated with specific locations [35], which can be leveraged to improve localization through map matching [36] or within a SLAM framework [37]. Certain activities, such as door opening, elevator button pressing, and card scanning, can be detected using smartwatches. In this paper, we focus solely on passage door opening due to the 2D scenario study and security considerations. To recognize this activity, a neural network combining a CNN and a gated recurrent unit (GRU) is used, enabling effective extraction of both spatial and temporal features for robust and accurate recognition of door-opening events. Once the door-opening activity is recognized, the position of the passage door from the floorplan is used to correct the user's location. Floorplans can be obtained via multiple channels. For public and commercial indoor environments (e.g., office buildings, shopping malls, and campuses), digital floorplans are

TABLE I
THE DEFINITION OF TYPICAL SYMBOLS

Symbol	Definition
$\boldsymbol{\omega}_k^b$	Angular velocity at the instant t_k in the body frame
$\mathbf{a}_k^b$	Specific force at the instant t_k in the body frame
$\boldsymbol{\omega}_k^w$	Angular velocity at the instant t_k in the world frame
$\mathbf{a}_k^w$	Specific force at the instant t_k in the world frame
${}_b^w\mathbf{R}_k$	Rotation matrix from the body frame to world frame at the instant t_k
$\widetilde{\mathbf{d}}_k$	Measured value at the instant t_k ($\mathbf{d}$ is a general symbol)
$\mathbf{d}_k^{ae}$	Variable related to attitude estimation at the instant t_k ($\mathbf{d}$ is a general symbol)
$\mathbf{d}_k^{pe}$	Variable related to position estimation at the instant t_k ($\mathbf{d}$ is a general symbol)
$\mathbf{d}_{k\|k-1}$	Predicted value in the filter at the instant t_k ($\mathbf{d}$ is a general symbol)

typically available from building information modeling (BIM) databases, facility management systems, or publicly released architectural drawings, from which door locations can be directly extracted. For scenarios without available digital floorplans, door locations and corridor layouts can be obtained via rapid indoor mapping using LiDAR-based or camera-based mobile mapping platforms.

Unlike map matching based on visual or point cloud data, map matching using human activity often suffers from high ambiguity, as it is difficult to infer a unique position from the same type of activity. Although activity sequence-based matching can enhance robustness, it relies on rich sequential data and degrades in performance when the activity sequence is sparse. To address this issue, a multi-hypothesis tracking within a linear system framework is formulated to integrate door-opening activity with WNIO. In the MHKF, the velocity in the world frame and its standard deviation are used to predict the position and its covariance. Once a door-opening activity is detected, multiple hypotheses are formed, each corresponding to a passage door in the floorplan and associated with a separate measurement model for position update. The weight of each hypothesis is then calculated to assess its confidence, and the one with the highest weight is selected as the final output.

Some key symbols used throughout this paper are defined in Table I. Two coordinate frames are considered: the body frame and the world frame. The body frame, denoted as b, is aligned with the three axes of the IMU. The world frame, denoted as w, originates from the body frame at the initial moment.

IV. SMARTWATCH-BASED NEURAL INERTIAL ODOMETRY

In traditional strapdown inertial navigation, the attitude is first estimated, and then the accelerometer data is transformed into the world frame. The velocity is then derived by integrating the transformed acceleration. This two-step integration process often results in increasing velocity errors, which in turn lead to rapid position drift. With the advent of deep learning, neural networks can bypass the integration process and directly regress more accurate velocity estimates by learning the underlying relationship between input features and target outputs.

In most data-driven inertial odometry approaches, raw IMU data is transformed from the body frame to the world frame to mitigate the effect of varying attitudes on the training data. However, if this front-end transformation is inaccurate, the neural network is forced to implicitly learn and compensate for these transformation errors, which typically leads to increased model complexity and the need for high-performance hardware, both impractical for smartwatch deployment. To address this issue, this paper focuses on improving the transformation accuracy by developing an adaptive ESKF-based attitude estimation module. It can mitigate the impact of external acceleration on attitude estimation, thereby eliminating the need for an overly complex network to learn nonlinear transformation errors.

A. Adaptive ESKF-Based Attitude Estimation

A pedestrian's arm tends to swing with varying amplitudes depending on the motion pattern. Although gyroscopes can track the arm's rapid changes in orientation, integrating their readings over time causes drift. Accelerometer data can be used to correct attitude estimation, but external linear accelerations during swings can significantly influence these estimates. Therefore, an adaptive scheme should be introduced to ensure accurate attitude estimation. An adaptive ESKF is formulated, and the state is defined as follows:

$$\mathbf{x}_k^{ae} = \begin{bmatrix} \mathbf{q}_k \\ \mathbf{b}_{g_k} \\ \mathbf{b}_{a_k} \end{bmatrix} \mathbf{q}_k \in \mathbb{R}^4, \mathbf{b}_{g_k} \in \mathbb{R}^3, \mathbf{b}_{a_k} \in \mathbb{R}^3 \tag{1}$$

where $\mathbf{x}_k^{ae}$ represents the state vector for attitude estimation at the instant t_k, comprising the quaternion, gyroscope bias, and accelerometer bias, following standard AHRS formulations. The error state can be represented as:

$$\delta\mathbf{x}_k^{ae} = \begin{bmatrix} \delta\boldsymbol{\theta}_k \\ \delta\mathbf{b}_{g_k} \\ \delta\mathbf{b}_{a_k} \end{bmatrix} \delta\boldsymbol{\theta}_k \in \mathbb{R}^3, \delta\mathbf{b}_{g_k} \in \mathbb{R}^3, \delta\mathbf{b}_{a_k} \in \mathbb{R}^3 \tag{2}$$

where $\delta\boldsymbol{\theta}_k$ is the rotation error, and $\delta\mathbf{b}_{g_k}$ and $\delta\mathbf{b}_{a_k}$ represent the gyroscope and accelerometer bias errors, respectively. Let the IMU measured angular velocity be:

$$\widetilde{\boldsymbol{\omega}}_k^b = \boldsymbol{\omega}_k^b + \mathbf{b}_{g_k} + \mathbf{n}_{g_k} \tag{3}$$

where $\boldsymbol{\omega}_k^b$ denotes the true angular velocity, and $\mathbf{n}_{g_k}$ denotes the gyroscope measurement noise.

The discrete-time propagation of the error state can be expressed as:

$$\delta\mathbf{x}_{k|k-1}^{ae} = \boldsymbol{\Phi}_{k-1}^{ae}\delta\mathbf{x}_{k-1}^{ae} + \mathbf{w}_{k-1}^{ae} \tag{4}$$

where $\delta\mathbf{x}_{k|k-1}^{ae}$ is the predicted error state at the instant t_k. $\boldsymbol{\Phi}_{k-1}^{ae} \in \mathbb{R}^{9\times 9}$ denotes the state transition matrix, which can be computed as:

$$\boldsymbol{\Phi}_{k-1}^{ae} = \exp\left(\mathbf{F}_{k-1}^{ae}\Delta t\right) \approx \mathbf{I} + \mathbf{F}_{k-1}^{ae}\Delta t \tag{5}$$

where Δt is the sampling interval of the IMU data. The matrix $\mathbf{F}_{k-1}^{ae}$ can be expressed as:

$$\mathbf{F}_{k-1}^{ae} = \begin{bmatrix} -\left[\widetilde{\boldsymbol{\omega}}_{k-1}^{b} - \mathbf{b}_{g_{k-1}}\right]_{\times} & -\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{6}$$

where the operator $[\cdot]_{\times}$ converts a vector into its skew-symmetric matrix representation. The process noise $\mathbf{w}_{k-1}^{ae}$ in equation (4) is assumed to be zero-mean Gaussian with covariance $\mathbf{Q}^{ae} \in \mathbb{R}^{9\times 9}$, $\mathbf{w}_{k-1}^{ae} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{ae})$. $\mathbf{Q}^{ae}$ can be given by:

$$\mathbf{Q}^{ae} = \begin{bmatrix} \sigma_g^2 \Delta t^2 \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \sigma_{b_g}^2 \Delta t \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \sigma_{b_a}^2 \Delta t \mathbf{I} \end{bmatrix} \tag{7}$$

where σ_g is the standard deviation of the gyroscope measurement noise, and σ_{b_g} and σ_{b_a} are the standard deviations of the gyroscope and accelerometer bias random walks, respectively. Then, the error covariance is propagated as:

$$\mathbf{P}_{k|k-1}^{ae} = \boldsymbol{\Phi}_{k-1}^{ae} \mathbf{P}_{k-1}^{ae} {\boldsymbol{\Phi}_{k-1}^{ae}}^{\mathrm{T}} + \mathbf{Q}^{ae} \tag{8}$$

where $\mathbf{P}_{k-1}^{ae}$ denotes the estimated covariance matrix at the instant t_{k-1}, and $\mathbf{P}_{k|k-1}^{ae}$ denotes the predicted covariance matrix at the instant t_k.

Assume that the specific force measured by the accelerometer is given by:

$$\widetilde{\mathbf{a}}_k^b = {}_b^w\mathbf{R}_k^{\mathrm{T}} \mathbf{g}^w + \mathbf{b}_{a_k} + \mathbf{n}_{a_k} \tag{9}$$

where ${}_b^w\mathbf{R}_k$ denotes the rotation matrix from the body frame to the world frame at the instant t_k. $\mathbf{g}^w$ represents the gravity vector in the world frame, and $\mathbf{n}_{a_k}$ is the accelerometer measurement noise.

Linearizing around the nominal state, the measurement model is:

$$\delta \mathbf{z}_k^{ae} = \mathbf{H}_k^{ae} \delta \mathbf{x}_k^{ae} + \mathbf{v}_k^{ae} \tag{10}$$

where $\mathbf{H}_k^{ae} \in \mathbb{R}^{3\times 9}$ denotes the measurement matrix for attitude estimation. $\mathbf{v}_k^{ae}$ represents the measurement noise for attitude estimation, which is derived from the accelerometer measurement noise $\mathbf{n}_{a_k}$. $\mathbf{H}_k^{ae}$ can be expressed as:

$$\mathbf{H}_k^{ae} = \begin{bmatrix} \left[{}_b^w\mathbf{R}_{k|k-1}^{\mathrm{T}} \mathbf{g}^w\right]_{\times} & \mathbf{0} & \mathbf{I} \end{bmatrix} \tag{11}$$

In the measurement update process, the Kalman gain $\mathbf{K}_k^{ae} \in \mathbb{R}^{9\times 3}$ can be calculated by:

$$\mathbf{K}_k^{ae} = \mathbf{P}_{k|k-1}^{ae} \mathbf{H}_k^{ae\mathrm{T}} \left(\mathbf{H}_k^{ae} \mathbf{P}_{k|k-1}^{ae} \mathbf{H}_k^{ae\mathrm{T}} + \mathbf{R}_k^{ae} \right)^{\mathrm{T}} \tag{12}$$

where $\mathbf{R}_k^{ae}$ denotes the measurement noise covariance, typically treated as constant. However, this fixed value cannot effectively handle external accelerations induced by arm movement, which can degrade attitude estimation. To mitigate this, we introduce an innovation-based adaptive scheme that dynamically updates the measurement covariance. The actual innovation is given by:

$$\mathbf{y}_k^{ae} = \mathbf{z}_k^{ae} - \mathbf{z}_{k|k-1}^{ae} = \widetilde{\mathbf{a}}_k^b - {}_b^w\mathbf{R}_{k|k-1}^{\mathrm{T}} \mathbf{g}^w - \mathbf{b}_{a_{k|k-1}} \tag{13}$$

where $\mathbf{z}_k^{ae}$ and $\mathbf{z}_{k|k-1}^{ae}$ denote the actual and predicted measurements, respectively. The theoretical innovation $\mathbf{S}_k \in \mathbb{R}^{3\times 3}$ can be expressed as:

$$\mathbf{S}_k = \mathbf{H}_k^{ae} \mathbf{P}_{k|k-1}^{ae} \mathbf{H}_k^{ae\mathrm{T}} + \mathbf{R}_k^{ae} \tag{14}$$

Then, the measurement covariance can be calculated by:

$$\mathbf{R}_k^{ae} = \frac{1}{N} \sum_{i=k-N+1}^{k} \mathbf{y}_i^{ae} \mathbf{y}_i^{ae\mathrm{T}} - \mathbf{H}_k^{ae} \mathbf{P}_{k|k-1}^{ae} \mathbf{H}_k^{ae\mathrm{T}} \tag{15}$$

To ensure positive definiteness, a small positive value ε can be added to diagonal elements of the estimated covariance matrix:

$$\mathbf{R}_k^{ae} = \mathbf{R}_k^{ae} + \varepsilon \mathbf{I} \tag{16}$$

The estimated covariance matrix will be used in equation (12) to obtain the Kalman gain. Then, the error state and covariance can be updated by:

$$\begin{aligned} \delta \mathbf{x}_k^{ae} &= \mathbf{K}_k^{ae} \left(\mathbf{z}_k^{ae} - \mathbf{z}_{k|k-1}^{ae} \right) \\ \mathbf{P}_k^{ae} &= \left(\mathbf{I} - \mathbf{K}_k^{ae} \mathbf{H}_k^{ae} \right) \mathbf{P}_{k|k-1}^{ae} \end{aligned} \tag{17}$$

Finally, the attitude $\mathbf{q}_k^{new}$ can be given by:

$$\begin{aligned} \mathbf{q}_k^{new} &= \delta \mathbf{q}_k \otimes \mathbf{q}_k \\ &\approx \begin{bmatrix} 1 \\ \frac{1}{2} \delta \boldsymbol{\theta}_k \end{bmatrix} \otimes \mathbf{q}_k \end{aligned} \tag{18}$$

B. CNN-Based Network Formulation

In this paper, we propose a lightweight CNN for velocity and uncertainty regression from smartwatch IMU data. The network is specifically designed to accommodate inertial signals while maintaining low complexity suitable for deployment on mobile devices. The overall architecture is shown in Fig. 2.

The network input is an IMU sequence. In Fig. 2, B, 6 and L represent the batch size, input feature dimension, and sequence length, respectively. Each sample consists of 3-axis accelerometer and 3-axis gyroscope measurements that are transformed from the body frame to the world frame over k time steps. The input is first passed through a lightweight 1D convolutional stem layer to project the raw signals into a higher-dimensional feature space suitable for temporal feature extraction. The core of the model is inspired by the architectural principles of MobileNetV4 [38], integrating a series of fused and depthwise separable inverted bottleneck blocks tailored for 1D time-series inertial data. These blocks are configured to progressively enhance representational capacity while preserving computational efficiency. Several later-stage blocks are augmented with squeeze-and-excitation (SE) modules, which dynamically recalibrate channel-wise responses to mitigate the impact of noisy or irrelevant sensor inputs, a common issue in wrist-mounted IMU data. Residual connections are employed when input and output dimensions align, promoting stable gradient propagation and temporal consistency.

To enrich nonlinear expressiveness without incurring substantial computational cost, HardSwish activation functions are used throughout the network. This choice strikes a balance between

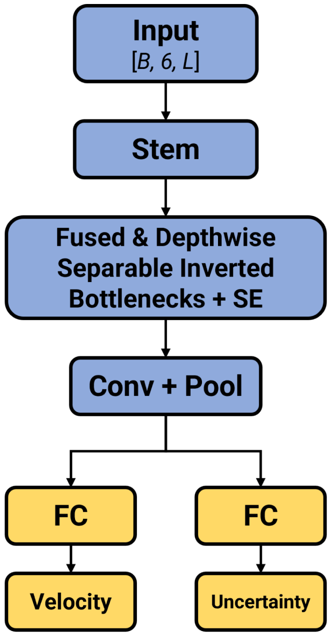


Fig. 2. The network structure of the smartwatch neural inertial odometry.

modeling capability and inference efficiency, particularly important for deployment on low-power edge devices. Following hierarchical feature extraction, a final pointwise convolution and global average pooling compress temporal features into a compact representation. Two parallel fully connected heads are used to regress the horizontal velocity vector and the logarithm of its standard deviation:

$$\mathbf{v}_k^w, \mathbf{u}_k^w = f\left(\boldsymbol{\omega}_{1:k}^w, \mathbf{a}_{1:k}^w\right) \tag{19}$$

The two components of $\mathbf{v}_k^w$ are denoted as $v_k^{w,x}$ and $v_k^{w,y}$, corresponding to the x- and y-axis velocities, respectively. The same notation is applied to $\mathbf{u}_k^w$, whose components are denoted as $u_k^{w,x}$ and $u_k^{w,y}$. The network is trained by minimizing a Gaussian negative log-likelihood loss:

$$\mathcal{L} = \frac{1}{B}\sum_{b=1}^{B}\sum_{j\in\{x,y\}}\left[\frac{\left(v_{k,b}^{w,j} - \hat{v}_{k,b}^{w,j}\right)^2}{2e^{2\hat{u}_{k,b}^{w,j}}} + \hat{u}_{k,b}^{w,j}\right] \tag{20}$$

where B is the batch size. The superscript w, j denotes the direction ($j \in \{x, y\}$) in the world frame. The hat symbol indicates predicted values, such as predicted velocity $\hat{v}_k^{w,j}$ and predicted log standard deviation $\hat{u}_k^{w,j}$.

C. Data Collection and Implementation Details

We use a dataset provided by Honor Device Co., Ltd., with all data collected using Honor's own smartwatch products. The complete dataset comprises 30 sequences collected from various individuals, primarily Honor staff. Each sequence contains approximately 30 groups of data, covering both left- and right-wrist wearing scenarios. The recorded motions include walking, jogging, and running. All data were collected in outdoor environments, as obtaining reliable ground truth labels for various motions in large-scale indoor spaces is challenging. The reference velocity was derived by applying a differential operation to high-accuracy positioning data obtained from a real-time kinematic (RTK) module mounted on the human body. The dataset is randomly divided into training and validation subsets, with 80% used for training and 20% for validation. It is important to note that the above data are used exclusively for training and validation purposes and are not involved in the testing phase. For testing, entirely unseen data collected by the author using a different smartwatch are employed.

During training, we apply data augmentation techniques to improve model robustness, including random horizontal rotations and perturbations on sensor bias and gravity direction. To simulate sensor bias, additive bias vectors are generated for each input sample, with each component independently sampled from a uniform distribution in the range of $\pm 0.4\ \mathrm{m/s^2}$ and $\pm 0.08\ \mathrm{rad/s}$. To perturb the gravity direction, samples are rotated along a random horizontal axis, with the rotation magnitude sampled uniformly from the range [0, 6] degrees. We trained the network using the Adam optimizer with an initial learning rate of 0.0001 and no weight decay. To mitigate overfitting, a dropout rate of 0.3 was applied before the fully connected heads. All training experiments were conducted on a single NVIDIA GeForce RTX 4090 D GPU.

V. Multi-Hypothesis Kalman Filter-Based Neural Inertial Localization

Even with accurate velocity prediction, inertial odometry inherently suffers from drift. In this section, we introduce the detection of door-opening activities combined with a floorplan to generate an absolute positioning solution, correcting drift in neural inertial odometry. To address ambiguity in matching, we construct a multi-hypothesis Kalman filter.

A. Door-Opening Activity Recognition

Since the smartwatch is worn on the wrist, interaction activities between the human arm and the environment can be more easily detected through the IMU. We analyze the IMU data and attitude during the door-opening process, using the Z-axis gyroscope data, Z-axis accelerometer data, and pitch as representative examples, as shown in Fig. 3. The grey area indicates the interval in which the door-opening activity occurs. It can be observed that the IMU data and attitude exhibit a clear difference compared to normal movement. Although this difference can be detected through sensor values, relying on them remains unreliable because it is difficult to select an appropriate threshold.

The combination of CNN and GRU is particularly effective. The CNN layers capture short-term motion patterns such as the rapid arm or body movements associated with pushing or pulling a door. Meanwhile, the GRU models longer-term temporal dependencies, including the sequence of approaching the door, initiating contact, and completing the opening motion. This hybrid design allows the model to recognize not just momentary gestures, but the entire temporal structure of the door-opening activity.

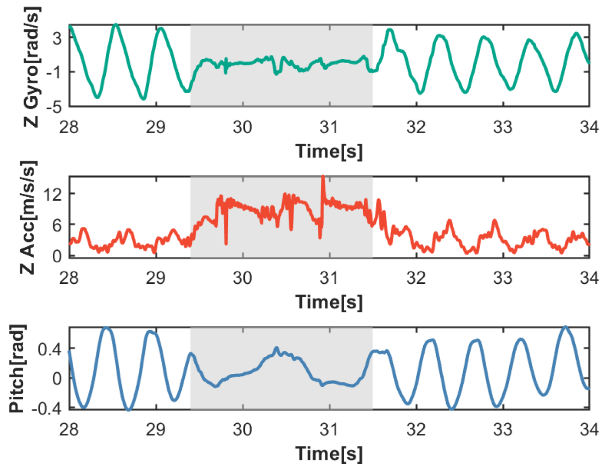


Fig. 3. The IMU data (Z-axis gyroscope and Z-axis accelerometer) and the pitch during normal movement and the door-opening process.

The model takes as input a segmented time-series tensor with shape $N \times C \times T$. Here, N denotes the batch size, $C = 8$ is the number of input channels, which include 3-axis gyroscope data, 3-axis accelerometer data, pitch, and roll, and T represents the window length. The window length is set to 200 samples, corresponding to 1.0 s at a 200 Hz IMU sampling rate. We adopt overlapping sliding windows, where each window shifts forward by 10 samples to maintain high temporal resolution. To avoid duplicate detections of the same action, once a door-opening activity is detected in a window, subsequent overlapping windows classified as door-opening are ignored, as they represent the continuation of the same event rather than a new one.

The network consists of two stages: a convolutional block for local feature extraction and a recurrent block for temporal modeling. In the convolutional block, the input sequence is first passed through a 1D convolutional layer that maps the input from C channels to 16 channels using a kernel size of 3, followed by batch normalization, ReLU activation, and max pooling to reduce the temporal resolution. A second 1D convolutional layer further transforms the representation to 32 channels, again followed by batch normalization and ReLU.

The output of the convolutional block is transposed and fed into a single-layer GRU with 64 hidden units. The GRU captures temporal dependencies in the time series and summarizes them into a final hidden state. This hidden state is then passed through a fully connected layer followed by a sigmoid activation function to produce a scalar output, which represents the predicted probability of a door-opening activity.

The model is trained using the binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(\hat{y}_i\right) + \left(1 - y_i\right)\log\left(1 - \hat{y}_i\right)\right] \tag{21}$$

where $y_i \in \{0, 1\}$ is the ground truth label for the i-th sample, and $\hat{y}_i \in (0, 1)$ is the predicted probability of the positive class.

We collected the training dataset ourselves. Five participants were asked to open different types of doors in our office buildings using the hand wearing the smartwatch. Each participant provided around 20 door-opening samples, resulting in a total of approximately 100 labeled door events. The IMU data from the smartwatch were recorded for door-opening activity recognition.

B. Multi-Hypothesis Kalman Filter

Multi-hypothesis approaches [39] have been widely used in state estimation, demonstrating strong performance especially in scenarios with data association uncertainty or ambiguous observations from multiple sources. Map matching based on door-opening activities inherently involves uncertain associations, as it is often impossible to directly determine the specific location corresponding to similar activity patterns. Therefore, multi-hypothesis tracking is particularly well-suited for addressing this problem.

The particle filter can be regarded as an implementation of the multi-hypothesis tracking paradigm, where each particle represents a possible state hypothesis. However, particle filtering is primarily designed for nonlinear and non-Gaussian systems. Moreover, it requires maintaining a large number of particles, which can significantly increase the computational burden on resource-constrained devices such as smartwatches. To address these challenges, we formulate a multi-hypothesis Kalman filter (MHKF) to integrate WNIO with activity-based map matching. On the one hand, our system model of position is linear, and the effects of non-Gaussian noise can be neglected. On the other hand, the MHKF offers improved efficiency. A branching structure is adopted to generate new hypotheses in response to ambiguous observations. To prevent the hypothesis set from growing excessively and to maintain computational efficiency, a pruning strategy is applied to retain only a small number of high-weight hypotheses.

The state vector for position estimation $\mathbf{x}_k^{pe} \in \mathbb{R}^2$ is defined as follows:

$$\mathbf{x}_k^{pe} = \begin{bmatrix} p_k^{w,x} \\ p_k^{w,y} \end{bmatrix} \tag{22}$$

where $p_k^{w,x}$ and $p_k^{w,y}$ represent the horizontal position in the world frame. Since the velocity regressed by the neural inertial odometry is expressed in the world frame, the attitude is not required in the state vector. The predicted state $\mathbf{x}_{k|k-1}^{pe}$ and covariance matrix $\mathbf{P}_{k|k-1}^{pe}$ at the instant t_k for position estimation are given by:

$$\begin{aligned} \mathbf{x}_{k|k-1}^{pe} &= \mathbf{x}_{k-1}^{pe} + \mathbf{v}_{k-1}^{w}\Delta T \\ \mathbf{P}_{k|k-1}^{pe} &= \mathbf{P}_{k-1}^{pe} + \mathbf{Q}_{k-1}^{pe} \end{aligned} \tag{23}$$

where $\mathbf{v}_{k-1}^{w}$ is the velocity regressed at the instant t_{k-1}, ΔT denotes the regression interval of the neural inertial odometry. $\mathbf{Q}_{k-1}^{pe}$ is the process-noise covariance induced by the regressed velocity, which can be expressed as:

$$\mathbf{Q}_{k-1}^{pe} = \mathrm{diag}\left(e^{2u_{k-1}^{w,x}}, e^{2u_{k-1}^{w,y}}\right) \tag{24}$$

The measurement model for position estimation is given by:

$$\mathbf{z}_k^{pe} = \mathbf{H}_k^{pe}\mathbf{x}_k^{pe} + \mathbf{v}_k^{pe} \tag{25}$$

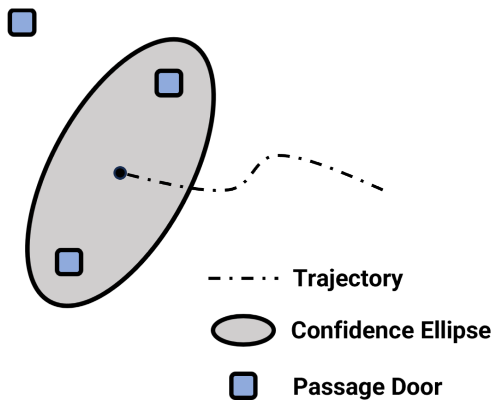


Fig. 4. The confidence ellipse and passage doors.

where $\mathbf{z}_k^{pe} \in \mathbb{R}^2$ denotes the passage door position obtained from the floorplan, $\mathbf{H}_k^{pe} \in \mathbb{R}^{2\times 2}$ represents the measurement matrix $\mathbf{I}$, and $\mathbf{v}_k^{pe}$ is the measurement noise for position estimation.

Each time a door-opening activity is recognized, different hypotheses need to be generated by associating the activity with different doors. Candidate doors are first selected based on the position uncertainty. The uncertainty of the user's position can be represented using a confidence ellipse. Assuming that $\mathbf{P}_{k|k-1}^{pe}$ at the instant t_k denotes the predicted covariance, it is expressed as follows:

$$\mathbf{P}_{k|k-1}^{pe} = \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \tag{26}$$

The radius of the confidence ellipse can be determined by:

$$\lambda_1 = \sqrt{\frac{P_{11}+P_{22}}{2} + \sqrt{\left(\frac{P_{11}-P_{22}}{2}\right)^2 + P_{12}^2}}$$
$$\lambda_2 = \sqrt{\frac{P_{11}+P_{22}}{2} - \sqrt{\left(\frac{P_{11}-P_{22}}{2}\right)^2 + P_{12}^2}} \tag{27}$$

where λ_1 and λ_2 denote the lengths of the major and minor axes, respectively. The angle α from the positive x-axis to the major axis of the ellipse can be expressed as:

$$\alpha = \begin{cases} 0 & P_{12} = 0 \; and \; P_{11} \geq P_{22} \\ \frac{\pi}{2} & P_{12} = 0 \; and \; P_{11} < P_{22} \\ atan2\left(\lambda_1^2 - P_{11}, \; P_{12}\right) & otherwise \end{cases} \tag{28}$$

According to equations (27) and (28), the ellipse can be described by the following equation:

$$x(t) = \sqrt{cs}\lambda_1 \cos\alpha \cos t - \sqrt{cs}\lambda_2 \sin\alpha \sin t + p_k^{w,x}$$
$$y(t) = \sqrt{cs}\lambda_1 \sin\alpha \cos t + \sqrt{cs}\lambda_2 \cos\alpha \sin t + p_k^{w,y} \tag{29}$$

where $t \in [0, 2\pi]$. cs stands for the chi-square value, which is set as 5.991in this paper with the confidence weight of 0.95. The relationship between the confidence ellipse of the position and the door position is given in Fig. 4. If a passage door lies within the range of the confidence ellipse, there is a certain weight that the pedestrian has passed through this door.

If there is only one door located within the confidence ellipse, the system directly uses that door's position to update the user's state. However, if two or more doors fall within the ellipse, the system creates a separate hypothesis for each candidate door. For each hypothesis, the likelihood of the current observation is computed based on the predicted state, assuming that door as the correct correspondence. For each hypothesis, the likelihood of the current observation is computed based on the predicted state $\mathbf{x}_{k|k-1}^{pe}$, which represents the prior estimate before incorporating the current measurement:

$$p\left(\mathbf{z}_k^{pe} | \mathbf{x}_{k|k-1}^{pe}\right) = \mathcal{N}\left(\mathbf{z}_k^{pe} | \mathbf{x}_{k|k-1}^{pe}, \mathbf{P}_{k|k-1}^{pe} + \mathbf{R}_k^{pe}\right) \tag{30}$$

where $\mathbf{R}_k^{pe}$ is the measurement noise covariance for position estimation. Then, the weight of each hypothesis is updated by:

$$\omega_i' = \omega_i p\left(\mathbf{z}_k^{pe} | \mathbf{x}_{k|k-1}^{pe}\right) \tag{31}$$

The updated weights are then normalized as:

$$\omega_i^{norm} = \frac{\omega_i'}{\sum_j \omega_j'} \tag{32}$$

where ω_i^{norm} is the normalized weight used for subsequent processing. Finally, the standard Kalman filter update is applied to each hypothesis individually, producing an updated state estimate for each branch. In addition, to prevent an unbounded increase in the number of hypotheses, we prune them by retaining only the top K hypotheses with the highest weights. Finally, the hypothesis with the highest weight is selected as the output.

VI. Experimental Evaluation

Real-world tests are conducted to validate the performance of the proposed method. Unlike the Honor smartwatch used for data collection during training, an Apple Watch Series 10 is used during testing. In addition, various movement paths with different motion patterns, including walking, jogging, and running, are evaluated. To further validate the impact of door-opening activities on localization, an office building with some passage doors is selected as the test site. In all experiments, the Livox Mid-360 is used to run FAST-LIO [40], which provides the ground truth. The experimental setup diagram is illustrated in Fig. 5.

A. Evaluation Methods and Metrics

As for the comparison, the following methods are included, which can be summarized as:

- *PDR-M [15]:* It estimates the pedestrian's location using model-based dead reckoning, where the Weinberg method [41] is employed to estimate step length, and PCA is used to determine walking direction.
- *PDR-CNN [32]:* This method employs a CNN model to estimate the 1D walking speed. For heading estimation, it utilizes the heading at each detected step to compute the heading change used in PDR.
- *WNIO:* This method is the proposed smartwatch-based neural inertial odometry.

TABLE II
ATE AND RTE COMPARISON UNDER DIFFERENT SCENARIOS

Method	Walking in the office building		Walking in the shopping mall		Running in the underground car park		Jogging in the underground car park	
	ATE (m)	RTE (m)	ATE (m)	RTE (m)	ATE (m)	RTE (m)	ATE (m)	RTE (m)
PDR-M	82.11	15.34	48.03	14.10	40.68	41.97	24.54	33.04
PDR-CNN	74.87	22.64	77.06	23.97	55.20	48.99	24.44	33.87
WNIO	**3.03**	**1.58**	**2.12**	**1.30**	**2.12**	**2.44**	**1.60**	**1.16**

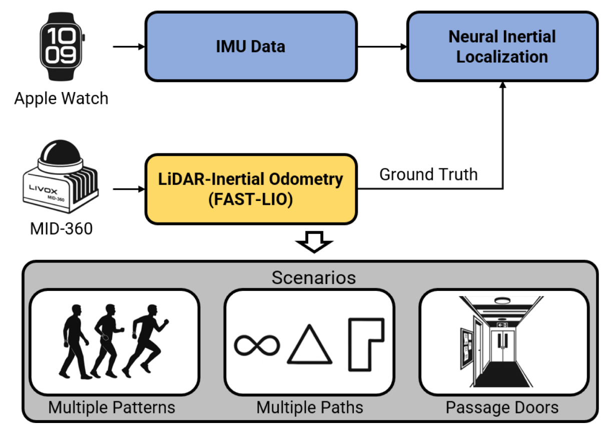


Fig. 5. The experimental setup.

- *Act-WNIO:* This method integrates WNIO with door-opening activity to enhance localization performance.

To qualify the positioning error, the absolute trajectory error (ATE) and relative trajectory error (RTE) are used. The detailed calculation method for ATE and RTE can be found in [42].

B. Experiment on Neural Inertial Odometry

The tests are first conducted to evaluate the performance of the proposed method in dead reckoning mode. Various paths with different motion patterns are included. Trajectory and positioning error comparisons among PDR-M, PDR-CNN, and WNIO under different scenarios are presented in Fig. 6 and Table II. In the office building experiments, we have included schematic maps for better visualization. However, the other scenarios, such as the underground car park and the shopping mall, are large open spaces where users walked freely without specific pathways. Therefore, schematic maps are not provided for those environments. In the semantic map, the grey boxes represent walls and rooms, while the red boxes indicate passage doors. It should be noted that in the neural inertial odometry experiments, door information is not used for position correction.

In Fig. 6, subfigures (a) and (b) represent walking modes in an office building and a shopping mall, respectively. In both cases, the walking distances are 293 meters and 243 meters, respectively, with an average walking speed of approximately 1.1 $\mathrm{m/s}$. It can be observed that WNIO achieves more accurate position estimation compared to PDR-M and PDR-CNN. PDR-M encounters an issue with reversed direction, whereas PDR-CNN suffers from an unknown misalignment relative to the actual movement trajectory. Although PCA can be utilized

TABLE III
THE POSITIONING ATE AND RTE COMPARISON BETWEEN ROTATED PDR-CNN AND WNIO IN THE OFFICE BUILDING

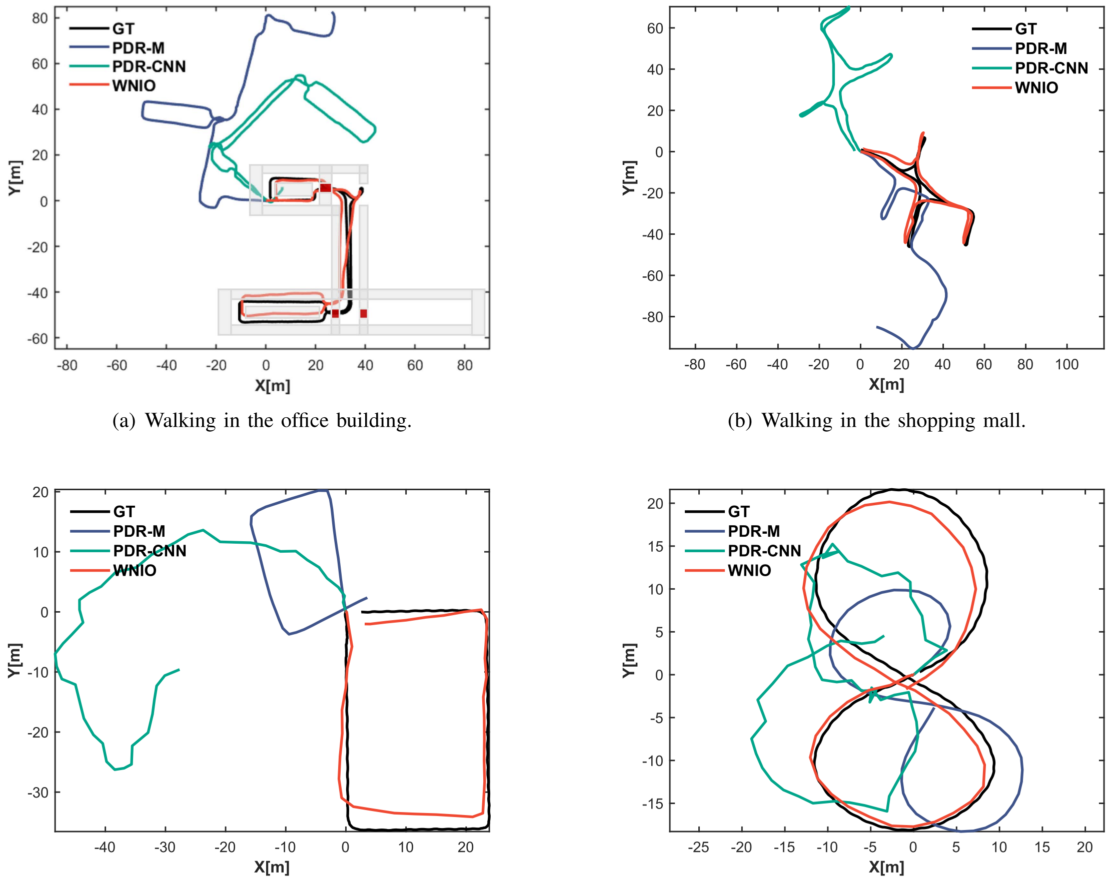
Method	ATE (m)	RTE (m)
PDR-CNN	4.71	1.99
WNIO	**3.03**	**1.58**

to estimate the walking direction, the inherent 180-degree ambiguity remains unresolved in smartwatch-based scenarios. As a result, PDR-M exhibits incorrect direction estimations at certain locations, leading to significant positioning errors. PDR-CNN determines heading change by using the heading at each detected step, enabling robust accumulation of relative orientation over time. However, since the heading accumulation starts from the wrist orientation at the initial step rather than the true walking direction, the resulting trajectory, although similar in shape to the ground truth, is subject to an unknown global rotation. This limits its performance in global frame alignment.

Subfigures (c) and (d) illustrate the running and jogging modes, respectively. The corresponding movement distances are approximately 113 meters and 121 meters, with average speeds of about 3.5 $\mathrm{m/s}$ and 2.5 $\mathrm{m/s}$. It can be observed that WNIO remains capable of tracking positions even in these challenging smartwatch scenarios. PDR-M continues to suffer from inaccurate heading estimation. Moreover, the scale issue becomes particularly evident in non-walking modes. During running or jogging, pedestrians typically exhibit longer step lengths. For model-based methods, estimating an appropriate step length is difficult without external information. As a result, the trajectories produced by PDR-M in non-walking modes appear shorter than the actual movement trajectories. Although the trajectory shape of PDR-CNN in walking mode is similar to the ground truth, its trajectory becomes difficult to recognize in running and jogging modes. This is mainly because the heading is derived from each detected step, and accurately estimating changes in the moving direction during these modes is more challenging.

In walking mode, PDR-CNN is primarily affected by the initial heading. In this part, we assume that the initial misalignment has been compensated. The trajectory and positioning error comparisons between the rotated PDR-CNN and WNIO in the office building are presented in Fig. 7 and Table III. It can be seen that WNIO still outperforms the rotated PDR-CNN. The heading estimation based on detected steps continues to introduce errors in the smartwatch scenario, resulting in deviated position estimates.

In addition, we compare the computational complexity of the proposed method with PDR-CNN, as shown in Table IV.



(a) Walking in the office building.

(b) Walking in the shopping mall.

(c) Running in the underground car park.

(d) Jogging in the underground car park.

Fig. 6. The trajectory comparison among PDR-M, PDR-CNN, and WNIO in different scenarios.

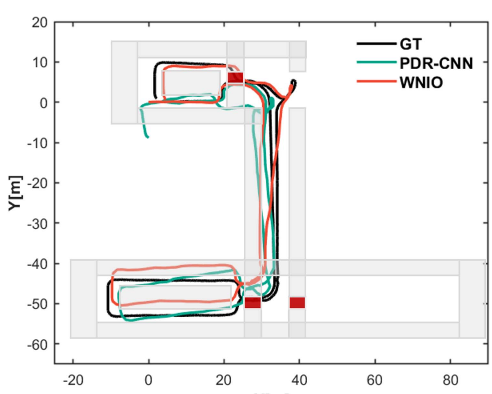


Fig. 7. The trajectory comparison between rotated PDR-CNN and WNIO in the office building.

TABLE IV
COMPUTATIONAL COMPLEXITY COMPARISON BETWEEN PDR-CNN AND WNIO

Network	MACs (M)	Parameters (M)
PDR-CNN (ResNet)	39.44	5.42
WNIO	**2.63**	**0.35**

PDR-CNN employs a ResNet-based architecture, a classical approach in CNN-based inertial odometry. The proposed model, however, uses depthwise separable convolutions along with a narrower and shallower architecture, reducing both operations and parameters. The comparison includes the number of multiply-accumulate operations (MACs) and parameters, where M denotes million. It can be observed that the proposed method achieves higher computational efficiency.

C. Experiment on Adaptive Attitude Estimation

In this paper, we employ an adaptive ESKF-based method to achieve accurate attitude tracking for constantly swinging arms. In this section, we discuss the importance of accurate attitude estimation in WNIO. We evaluate two variants of the proposed WNIO system, both of which use the same back-end neural network as WNIO, but differ in their front-end attitude estimation modules.

The first variant, WNIO(NAE), uses a conventional ESKF-based method without the adaptive mechanism. The second variant, WNIO(AMA), incorporates an adaptive Madgwick filter [43] for attitude estimation. Specifically, the adaptive Madgwick filter operates at 200 Hz. The gain factor β is adjusted based on the deviation of the measured acceleration magnitude

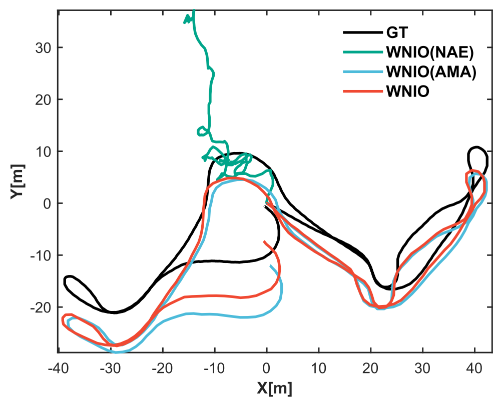


Fig. 8. The trajectory comparison among WNIO(NAE), WNIO(AMA), and WNIO in the shopping mall.

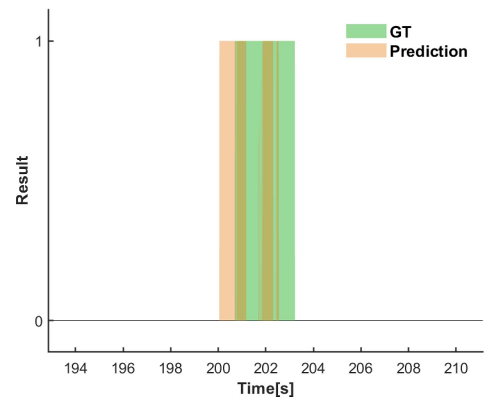


Fig. 10. The comparison between predicted detection results and ground-truth door-opening labels.

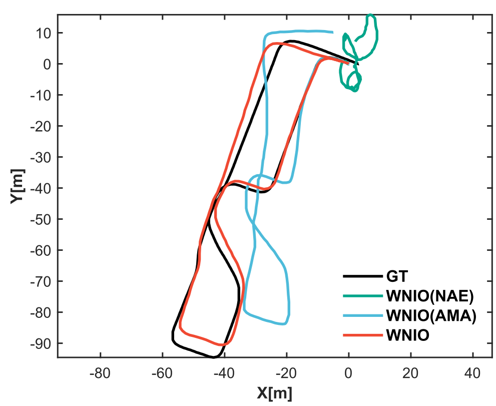


Fig. 9. The trajectory comparison among WNIO(NAE), WNIO(AMA), and WNIO in the underground car park.

TABLE V
THE POSITIONING ATE AND RTE COMPARISON BETWEEN WNIO(NAE), WNIO(AMA) AND WNIO IN THE SHOPPING MALL AND UNDERGROUND CAR PARK

Method	Shopping mall		Underground car park	
	ATE (m)	**RTE** (m)	**ATE** (m)	**RTE** (m)
WNIO(NAE)	35.35	14.67	64.10	14.18
WNIO(AMA)	6.36	1.27	14.13	3.85
WNIO	**5.12**	**1.12**	**3.23**	**1.07**

from the gravity norm. When the acceleration is close to gravity, indicating low dynamic motion and a reliable reference, a larger β (0.08-0.2) is used to effectively suppress gyroscope drift. Conversely, when the arm motion is highly dynamic and the accelerometer contains significant linear accelerations, β is reduced (0.01-0.04) to avoid introducing motion-induced errors into the attitude estimation. Experiments are conducted at two different sites: a shopping mall and an underground car park, with total walking distances of approximately 252 meters and 274 meters, respectively. The trajectory and positioning error comparisons are presented in Fig. 8, Fig. 9, and Table V.

In WNIO(NAE), the accelerometer data is directly used to aid attitude estimation under all motion conditions. This can lead to significant estimation errors when external acceleration is present. As a result, the neural network may regress incorrect velocities from the erroneous acceleration in the world frame, leading to unusable position estimates. Although WNIO(AMA) performs better than WNIO(NAE), it is still affected by the continuously swinging motion, resulting in inaccurate position estimates. This is primarily because the filter lacks an explicit model to distinguish gravity from linear acceleration. In contrast, WNIO uses an adaptive ESKF that leverages innovation-based statistical inference to dynamically adjust the trust in measurements. This enables more robust and accurate attitude estimation, ultimately leading to improved position accuracy.

D. Experiments on Activity-Aided Inertial Localization

From the above sections, it can be seen that although WNIO achieves better position tracking than traditional PDR methods, it still suffers from drift. In this section, we evaluate the effectiveness of incorporating door-opening activities to constrain drift in WNIO. We conducted experiments at a site containing three passage doors, with a total moving distance of approximately 370 meters.

The comparison between the predicted detection results and the ground-truth door-opening labels at a specific door is shown in Fig. 10. It can be observed that the predicted door-opening activities are mostly concentrated around the period of the ground-truth labels. However, some instances remain undetected. It is worth noting that door-opening is a continuous activity, while our method performs classification over overlapping windows. As a result, some windows near an actual door-opening sequence may be labeled as negative. Nevertheless, for localization purposes, detecting at least one positive window in close proximity to the action period is sufficient to confirm the event and apply position correction.

The trajectory and positioning RMSE comparisons between WNIO and Act-WNIO are presented in Fig. 11 and Table VI, respectively. The door locations, indicated by the red boxes, are used for position correction in this test. It can be observed that WNIO exhibits noticeable drift after prolonged movement, which is an inherent limitation of dead reckoning.

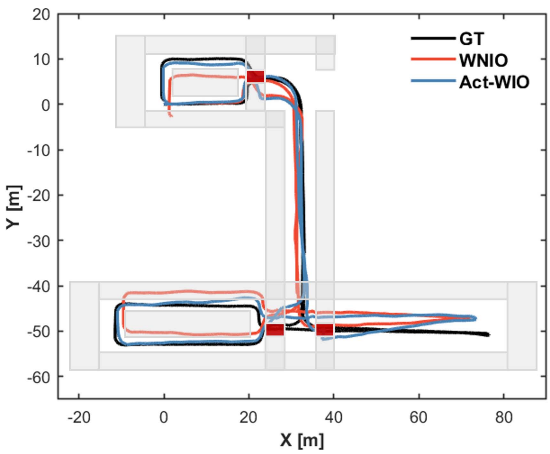


Fig. 11. The trajectory comparison between WNIO and Act-WNIO in the office building.

TABLE VI
THE POSITIONING RMSE COMPARISON BETWEEN WNIO AND ACT-WNIO IN THE OFFICE BUILDING

Method	**RMSE** (m)
WNIO	3.38
Act-WNIO	**2.82**

However, by incorporating floorplan information and known door positions, our proposed method enables door matching via smartwatch-based activity detection, effectively mitigating error accumulation. The doors used for testing are different from those used for training, enabling evaluation on previously unseen data. The results show that our method can successfully recognize door-opening activities without requiring samples from specific doors in each scenario. While additional data collection could further improve scalability, only fine-tuning is needed rather than full retraining. When deploying in a new environment, only the floor plan and door locations are required, without any additional labeled activity data. In addition, the proposed method mainly detects door-opening activities performed with the hand wearing the smartwatch, since the smartwatch IMU data directly reflect the corresponding arm motion. When users open a door with the non-smartwatch hand, relying solely on the smartwatch IMU data makes the activity recognition more challenging. Nevertheless, in our field tests, users also opened doors using their non-smartwatch hand. In such cases, position correction is not triggered. Even so, the localization performance still benefits from the limited number of detected door events when the smartwatch-wearing hand is used.

It is important to note that, due to safety concerns and the 2D nature of the scenarios, only passage door-opening activities are used in this paper to correct the position. However, if conditions permit, additional activities, such as card scanning, taking elevators or escalators, and using stairs, can be integrated into the proposed method to extend its applicability to larger-scale scenarios.

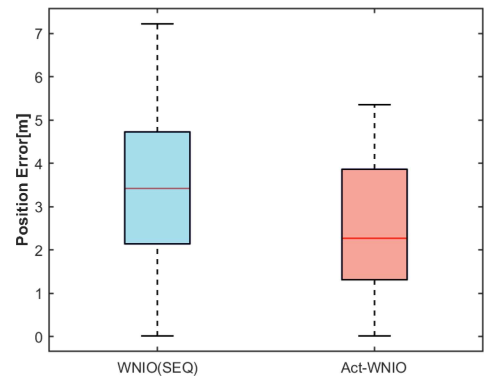


Fig. 12. The positioning error comparison between WNIO(SEQ) and Act-WNIO in the office building.

In addition, we compare the performance of the proposed Act-WNIO with the activity sequence-based matching method, denoted as WNIO(SEQ). Fig. 12 shows the positioning error comparison between WNIO(SEQ) and Act-WNIO in the office building. It can be seen that the proposed method still achieves better performance. The activity sequence-based method struggles to promptly correct the position when the activity sequence is sparse. In the test, the sequence number was set to 2, and the duration between the first and second activities was long, causing the position estimate of WNIO(SEQ) to exhibit significant drift before the second activity was detected. Moreover, compared with WNIO(SEQ), MHKF can adaptively compute the optimal gain based on the predicted state covariance and the observation covariance. This allows it to achieve the minimum mean squared error, thereby leading to better positioning performance.

E. Experiments on Differentiation Among Similar Activities

Many common human activities, such as opening windows, cabinets, or even gestures like waving, involve motions that are kinematically similar to door opening, which may adversely affect the proposed method. Nevertheless, noticeable differences can still be observed in their IMU data. For example, the hand is typically raised higher during waving than during door opening. In addition, the structural and operational differences between door and window handles give rise to distinguishable IMU signal signatures. As illustrated in Fig. 13, the Z-axis accelerometer data during window-opening and door-opening processes exhibit different local and temporal features. These differences can be exploited by the CNN and GRU to differentiate among similar activities.

We conducted experiments involving door-opening, window-opening, and waving activities using the proposed method. The results are shown in Fig. 14, where the upper subfigure (a) illustrates the performed activities and the lower subfigure (b) presents the corresponding detection results. The green blocks indicate the actual door-opening activities, while the red and blue blocks represent the actual window-opening and waving activities, respectively. The orange blocks denote the detected activities. It can be observed that only door-opening activities

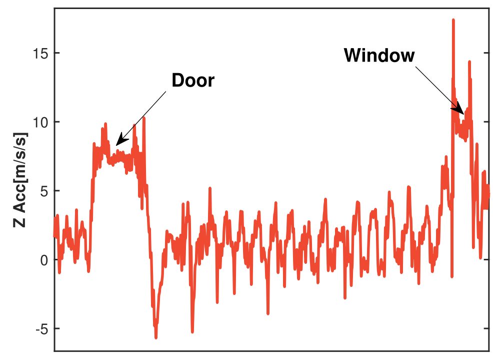


Fig. 13. The Z-axis accelerometer data during the window-opening and door-opening processes.

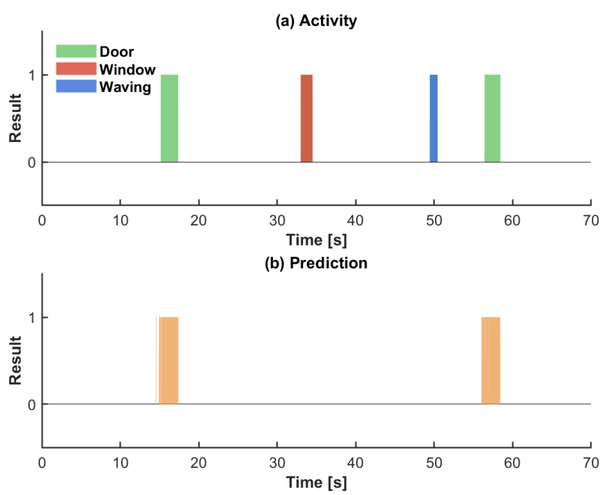


Fig. 14. The comparison between the predicted detection results and the ground-truth activity labels.

are detected, without misclassifying window-opening or waving actions as door opening.

However, the generalizability of the proposed model remains a challenge, as it cannot be guaranteed that the current model can handle all possible corner cases. Misclassification may occur when users perform complex motions. Increasing the amount of training data is an effective approach, however, it incurs substantial costs in data collection. To improve the practicality and robustness of the method, several constraints are incorporated into the detection framework. First, the distance between the user and the door is used to determine whether the user is in the vicinity of a door. If the estimated distance is within a predefined threshold (e.g., 6 meters), the detected activity is more likely to correspond to a door-opening activity. On this basis, the relationship between the estimated user position and the room layout is further utilized as a reference. If the estimated positions indicate that the user is transitioning between a corridor and a room (local PDR results are reliable), the detected activity is considered a door-opening activity. Otherwise, the detected activity will not be used as similar motions may be caused by interactions with windows, cabinets, or other objects.

VII. Conclusion

This paper presents a neural inertial localization method based on a single smartwatch. The proposed approach, referred to as WNIO, incorporates a physics-guided attitude estimation module to provide robust and interpretable orientation, even in challenging smartwatch scenarios caused by complex arm motion. This pre-alignment simplifies the downstream learning task and improves generalization across diverse motion patterns, including walking, jogging, and running, enabling more accurate and reliable dead reckoning performance compared with traditional methods. Furthermore, semantic human activities recognized by the smartwatch, such as the door-opening activity, are leveraged to constrain error accumulation and enhance long-term localization accuracy. A multi-hypothesis Kalman filter is developed for position matching, achieving better performance than traditional activity-sequence-based matching methods. In future work, we plan to integrate external proximity information, such as near-field communication (NFC) and BLE, to provide additional absolute position corrections and enable a more seamless and general indoor localization solution.

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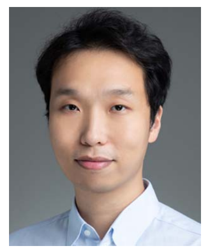

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